wellcome connecting science

Antimicrobial Resistance of Bacterial Pathogens – Africa

End to end challenge

XDR Typhoid hits the AMR course

March 2026

End to End Challenge

AMERICAN SOCIETY FOR MICROBIOLOGY mBio®

RESEARCH ARTICLE

Check for updates

Emergence of an Extensively Drug-Resistant *Salmonella enterica* Serovar Typhi Clone Harboring a Promiscuous Plasmid Encoding Resistance to Fluoroquinolones and Third-Generation Cephalosporins

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The challenge

- Today you are going to work at pace to understand the emergence and spread of an XDR Typhoid lineage in ~~Pakistan~~ Campus!

- Since November 2016, a large proportion of ceftriaxone-resistant cases have been identified in the province of Sindh, Pakistan, primarily from the cities of Hyderabad and Karachi.

- You will be given the sample metadata and AMR profiles and raw sequencing data (fastq files)

- You will then process the genomes (though we have provided the outputs at various stages) to:

  - Assembly and identify presence of resistance genes – Using Pathogenwatch and AMR finder
  - Compare the XDR strain to a WT reference genome to explore the acquisition of resistance
  - Map the genomes and build phylogenies
  - Integrate the resistance data into the phylogenies using Microreact to understand the source and evolution of the outbreak

- **https://five.epicollect.net/project/south-africa-typhoid-xdr-outbreak**

- To collect epidemiological and AST data on your assigned S. typhi sample using Epicollect
- To run the de novo assembly on your assigned *S. typhi* genome
- To run Snippy on your assigned *S. typhi* genome to obtain consensus sequence
- To run a phylogenetic analysis of all 24 S. typhi cases investigated plus contextual samples
- To detect AMR genetic markers from the assembly of your assigned sample using Pathogenwatch and AMRFinderPlus
- To investigate the context of AMR genes and MGEs using Artemis & ACT
- To obtain the MLST and cgMLST of your assigned sample
- To investigate the origin of the XDR S. typhoid outbreak using MicroReact

- https://github.com/WCSCourses/AMR_2026/tree/main/course_modules_2026/End-to-End_challenge
- **https://five.epicollect.net/project/south-africa-typhoid-xdr-outbreak**

Wellcome Sanger Institute 4-year PhD programme

Annual – closes end of November each year

Students offered one of our 4-year PhD programme funded studentships will obtain full financial support, including University tuition fees, regardless of nationality.

There are 12 funded studentships available across all our research areas. Any students with their own external studentship funding are still expected to apply to the 4-year PhD Programme in the usual way.

https://www.sanger.ac.uk/about/study/phd-programmes/4-year-phd-programme/

https://www.sanger.ac.uk/programme/parasites-and-microbes/

MPhil in Genomic Science

In 2016, the Institute started a new 1-year research MPhil programme that aims to equip students from a number of low and middle income countries with a unique blend of experimental and informatics skills, maximising their competitiveness for future opportunities. The programme is run in partnership with the following institutions **only:**

- **The KEMRI-Wellcome Trust Research Programme, Kenya**
- **The Malawi-Liverpool-Wellcome Trust Clinical Research Programme**
- **The Africa Health Research Institute, South Africa**
- The Mahidol-Oxford Tropical Medicine Research Unit, Thailand and Laos
- The Oxford University Clinical Research Unit, Vietnam
- The Oxford University Clinical Research Unit, Nepal
- **MRC Unit The Gambia at LSHTM**
- **MRC/UVRI & LSHTM Uganda Research Unit**
- **KEMRI Nairobi, Kenya**
- **University of Science, Techniques and Technologies of Bamako, Mali**
- International Centre for Diarrhoeal Disease Research, Bangladesh
- **Ifakara Health Institute, Tanzania**
- Universidad de Antioquia, Colombia
- Universidad Regional Amazonica Ikiam, Ecuador

- The institutions of the current and former Sanger Institute International Fellows

The programme funds (stipend and University tuition fees) up to 4 students per year and applicants must be nominated by one of the partner institutions. Assessment of the MPhil is by submission of a thesis and an oral examination. Students start on the 1 October, with submission of the thesis by the end of August

Gates Cambridge - https://www.gatescambridge.org/programme/the-scholarship/

https://www.cambridge-africa.cam.ac.uk